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Discovering a Defensible 5G/6G Research Problem: Checkpoint 1 Report on Certified Collateral-Risk Control for Slice-Aware Intrusion Response

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Abstract. We report Checkpoint 1 of an independent research-discovery mission whose goal is a novel, feasible, and defensible IEEE Transactions-level problem in 5G/6G networks. We screened 75 sources across ten research clusters; 72 papers and preprints are usable, each recorded with verification status and provenance in a reusable structured bibliography. From them we generated eleven candidate problems, scored them with a fixed weighted rubric, and applied mandatory rejection criteria. Six candidates were rejected and five retained; the top three went to an adversarial novelty audit. The audit shows that detection-level conformal false-alarm control for intrusion detection, the direction closest to the team's prior work, became crowded in 2025-2026, so we reject it as a primary topic. We provisionally recommend certified collateral-risk control for slice-aware intrusion response: a multi-action mitigation policy calibrated, through multiple-testing risk control, so that the oracle-counterfactual SLA-violation risk imposed on benign users of each slice stays within a slice-specific budget with a finite-sample, distribution-free guarantee. Our backup is SLA-weighted multi-slice quickest attack detection. The report states falsifiable hypotheses, baselines, and minimum evidence, and lists the access gaps that block final novelty claims. It contains no experimental results.

Keywords: 5G network slicing, intrusion response, conformal risk control, learn-then-test, service-level agreements, research methodology

1 Introduction

This report is **Checkpoint 1** of an independent research-discovery mission. The mission asks for a novel, feasible, and defensible IEEE Transactions-level research problem in 5G/6G systems. Its deliverable is a validated research direction and an executable plan, not a manuscript with results. As the brief requires, the report stops after the initial checkpoint (Stages A–F, preliminary forms only). It contains **no experimental results and no performance numbers** for any proposed method. Every claim is tagged with one of four evidence levels:

- **[Verified]**: confirmed in a retrieved bibliographic or repository record.
- **[Inference]**: an evidence-supported inference by the authors of this report.
- **[Hypothesis]**: a falsifiable statement that has not yet been tested.
- **[Unverified]**: could not be checked in this session.

Summary of the provisional outcome. We screened 75 sources across ten 5G/6G research areas. Of these, 72 papers and preprints are usable; 3 entries are excluded from that count (2 software artefacts and 1 unconfirmed preprint). From them we generated eleven candidate problems. Six were rejected under the mandatory criteria and five retained; the three highest-scoring retained candidates (C1, C3, C6) went to an adversarial novelty audit. The most important negative finding is this: detection-level conformal false-alarm control for intrusion detection, the direction that best matched the team’s prior background, has become crowded in 2025–2026 [1–5]. It is therefore **rejected** as a primary topic.

The provisional primary recommendation moves the guarantee one layer down the control loop, from alerts to the service-level consequences of mitigation actions:

> **C1 — Risk-controlled, slice-aware intrusion response.** Calibrate detector-triggered mitigation policies (rate-limit, quarantine to a sinkhole slice, block) for a sliced 5G core. The calibration should come with a finite-sample, distribution-free guarantee that the collateral SLA-violation risk imposed on benign users of each slice stays below a slice-specific budget, while residual attack impact is minimised.

The backup is **C3**, SLA-weighted multi-slice quickest attack detection. Both recommendations are explicitly **provisional**. Section 9 lists the full-text reads and infrastructure checks that must pass before Phase 2.

2 Methodology of this checkpoint

Search protocol. We ran targeted web searches (about 60 queries, standard and extended) on 2 November 2026. They covered ten topic clusters from the brief (Section 3), plus explicit adversarial searches for each shortlisted claim (Section 6). For each source we recorded the citation, venue, year, DOI or preprint identifier, the problem, contribution, method, environment, findings, limitations, relevance, and verification status. The structured bibliography is `data/bibliography.json`. The matrix (`data/literature_matrix.csv`) and the BibTeX file are generated from it by `scripts/build_bib.py`.

Verification levels. Each entry is labelled V-DOI, V-META, V-PRE, PARTIAL, or UNVERIFIED (Table 1). **No DOI was guessed.** Where a DOI did not appear in any retrieved record, the field is left blank.

Hard constraints of this session [Verified]. The container's network policy blocked arxiv.org, doi.org, api.crossref.org, OpenAlex, and Semantic Scholar (HTTP 403 or DNS failure). Only a search-engine interface was available. As a result:

1. **No full text was read.** Every entry was reviewed at the abstract and metadata level. Limitations marked [A] in the matrix are the authors' own statements; those marked [I] are our inferences and must be confirmed against the full text.
2. DOIs could not be resolved independently.
3. The repository supplied for this mission was **empty**: no prior manuscripts, code, configurations, logs, or datasets. Claims about the team's prior work and testbed therefore stay **[Unverified]**.

Verification status of the structured bibliography.

Level	Meaning	Count
V-DOI	Title, authors, venue, year, and DOI seen together in a retrieved record	24
V-META	Metadata verified; DOI not seen (left blank)	20
V-PRE	Preprint (or software artefact) exists; no peer-reviewed version confirmed	19
PARTIAL	Exists; at least one metadata field conflicts or is unconfirmed	11
UNVERIFIED	Seen once; must not be cited until checked	1
Total		**75**

Level	Meaning	Count
	45 of the 75 entries are dated 2023–2026; 7 are surveys used only as background	

Honest assessment of the minimum literature target. The brief asks for at least 60 verified relevant papers. We have 72 usable papers and preprints, so the count is met. The depth of verification is weaker than a Transactions-grade review requires: only 24 entries have a confirmed DOI, 19 are preprints, and none was read in full. We therefore treat the landscape as **sufficient for screening and ranking, but not for final novelty claims**. Phase 1 of the roadmap (Section 8) closes this gap.

3 Stage A — Research landscape

Table 2 organises the evidence into ten clusters. The full per-paper matrix (75 rows × 19 fields) is supplied as `data/literature_matrix.csv`.

Structured research landscape.

#	Cluster	Representative verified works	State of the art (evidence level)	Open space
L1	Validity of ML-NIDS evaluation	[6–12]	Benchmark results are inflated by sampling bias, temporal leakage, label errors, and shortcut features. There is a broad consensus that false-positive cost, not accuracy, blocks deployment. [Verified, abstract level]	Operational, cost-aware evaluation is still rare.
L2	Cross-domain generalisation	[13–15]	Detectors degrade sharply across datasets; NetFlow feature standardisation enables cross-testing. [Verified]	No 5G slice-aware cross-domain study found. [Inference]
L3	5G-specific security data and detection	[16–22]	Detection dominates. Reported accuracies are near-perfect (for example 99.99% in [18], per its abstract). SAGE-5GC shows adversarial fragility. [Verified]	**Response and mitigation, and their service-level side effects, are almost absent.** [Inference]
L4	Drift and conformal security	[1–5, 23–26]	2025–2026 saw a surge of conformal and risk-calibrated IDS: FPR or FDR control, regime-dependent CRC, robustness to adversarial orbits. [Verified, preprint level]	Detection-level guarantees are now crowded; action-level guarantees are not. [Inference]
L5	Distribution-free risk-control theory	[27–36]	Mature tools exist for monotone (CRC) and non-monotone or multiple (LTT) risks, covariate shift, and online shift. Action gating with risk control is ap-	Not yet applied to coupled, multi-action network mitigation, as far as found. [Inference]

#	Cluster	Representative verified works	State of the art (evidence level)	Open space
			pearing in other domains (GUI agents). [Verified]	
L6	Intrusion response and mitigation	[37–43]	Model-based optimal response (POMDPs, stochastic games, optimal stopping) yields threshold policies; line-rate DDoS mitigation exists in P4. [Verified]	No distribution-free guarantee on collateral harm; no slice awareness. [Inference]
L7	Slicing and SLA-aware resource management	[44–49]	Asymmetric SLA-violation costs and overbooking create cross-slice coupling. [Verified]	Coupling between security actions and SLAs is unexplored. [Inference]
L8	O-RAN and AI-native RAN control	[50–57]	xApp development, explainability, conflict management (PACIFISTA, TMC 2025), interface security. [Verified]	All require RAN or RIC infrastructure that is not available here.
L9	Adversarial attacks on learned slicing	[58–61]	Jamming and flooding against DRL slicing cause slice-dependent SLA violations. [Verified, preprint level]	Crowded, and it needs PHY access.
L10	Digital twins, energy, FL, LLM, QCD	[62–71]	GNN twins give point predictions. Energy work needs RAN measurements. FL-IDS and LLM-ops are crowded or hype-driven. Multistream QCD theory is mature. [Verified]	Calibrated twins (C6) and SLA-weighted multi-slice QCD (C3) are moderately open. [Inference]

Cross-cutting observations.

1. **[Inference]** The 5G security literature is stuck at the detection stage on near-saturated benchmarks. Clusters L1 and L4 already explain why accuracy claims are unreliable.
2. **[Verified]** Mature distribution-free risk-control tools (L5) exist, and conformal IDS work (L4) uses them, but only on alert-level quantities such as FPR, FDR, and alert budgets.
3. **[Inference]** The service-level consequence of acting on alerts is what operators pay for in a sliced network, and especially in URLLC. It is governed by model-based response theory (L6) that offers no finite-sample guarantee under model misspecification. It is also coupled across slices through shared resources (L7).

The intersection of points 2 and 3 is the gap pursued by C1.

Coverage limitations. PHY-layer and IEEE TWC-style original research (beamforming, ISAC, RIS) is under-represented. It was screened out early because nothing indicates that the team has PHY or SDR infrastructure (Section 7). This is a deliberate scoping decision, recorded in the decision log as D-04, not evidence that PHY topics are less important.

4 Stage B — Candidate research problems

We generated eleven candidates, one more than the ten required, so that the rejected C11 could be documented. Each is summarised along the brief's five dimensions: A (problem), B (novelty), C (technical depth), D (feasibility), and E (evaluation). The shortlisted candidates are expanded in Sections 6 and 8.

C1 — Risk-controlled, slice-aware intrusion response.

- A. Mitigation actions taken on imperfect detections (rate-limiting, quarantine, blocking) harm benign users. In a sliced core this harm propagates through shared UPF and transport resources, and its cost differs sharply by slice (URLLC versus mMTC). Current IDS work bounds alert-level error but not the SLA damage of the actions taken.
- B. Closest works: CALIBURN [1] (FP-budgeted alerts), FIRCE [2] ("response" means retraining), CORA [35] (risk-controlled action gating, non-network), Hammar & Stadler [37] (model-based optimal response), and SliceSecure [18] (slice detection only). **Missing piece [Inference]:** a distribution-free, per-slice guarantee on the collateral SLA risk of a multi-action mitigation policy under cross-slice coupling.
- C. Multi-risk LTT calibration over a structured policy space, counterfactual loss evaluation with a twin-error propagation bound, and robustness to temporal dependence and shift.
- D. Needs an emulated sliced 5G core (Open5GS or free5GC with UERANSIM), a traffic and attack harness, and KPI measurement. Moderate to high complexity. Main risk: the cost of counterfactual replay.
- E. See Section 8.

C2 — Slice-stratified conformal false-alarm control for 5G NIDS (detection only).

- A. Bound the per-slice FPR of a detector.
- B. **Contradicted.** Marginal FPR and FDR control [31], conformal IDS [2, 3, 25], regime-dependent CRC [1], and online conformal FDR [5] already exist. Stratifying by slice is a Mondrian-style re-grouping and too thin for a Transactions paper.
- C. Low.
- D. High feasibility.
- E. Straightforward.
- **Rejected:** gap contradicted.

C3 — SLA-weighted multi-slice quickest attack detection.

- A. Detect attack onset across slice streams so as to minimise SLA-weighted detection delay subject to heterogeneous per-slice false-alarm constraints, with streams coupled by shared resources.
- B. Closest works: [69, 70], plus an unpublished simulation [71]. **Missing piece [Inference]:** heterogeneous delay weights and false-alarm constraints with coupled streams.
- C. Asymptotic optimality analysis is possible and necessary.
- D. Trace-driven evaluation is feasible.
- E. Delay versus ARL curves against CUSUM and SUM-CUSUM variants.
- **Risk:** the result may be a corollary of the existing multichannel theory.

C4 — Leakage-audited cross-testbed benchmark for 5G NIDS.

- **Rejected.** It probably overlaps the team's own leakage-audit work [Unverified: prior work not provided], and a 5G-NIDD realistic-split package already exists [72]. Valuable as infrastructure for C1, not as the contribution.

C5 — Conflict-aware xApp composition with guarantees.

- **Rejected.** PACIFISTA [55] covers conflict detection and severity, and no RAN or RIC infrastructure has been verified.

C6 — Calibrated-uncertainty GNN digital twin for slice SLA prediction.

- A. GNN twins [62] give point estimates; operators need calibrated SLA-violation risk.
- B. No conformal-twin work was found in this session's searches, but **the search was shallow [Unverified]**.
- C. Conformal regression over graph-structured, exchangeability-breaking data.
- D. Feasible on public RouteNet datasets, but the 5G relevance is indirect (wired topologies).
- E. Coverage and width against Bayesian and ensemble GNNs.

C7 — Adversarial robustness of DRL RAN-slicing controllers.

- **Rejected.** It needs PHY, RAN, or SDR access, and the topic is crowded [58, 60, 61].

C8 — Uncertainty-aware RAN sleep-mode control.

- **Rejected.** It needs RAN energy measurements [64]; in simulation alone it cannot support operational claims.

C9 — Federated slice-level IDS under non-IID data.

- **Rejected.** Crowded [65, 66], with no identifiable new contribution.

C10 — Adversarially robust 5GC PFCP anomaly detection.

- Retained but not shortlisted. SAGE-5GC [20] has just defined the evaluation protocol; a defence paper would be incremental unless it adds certified robustness [4].

C11 — LLM-driven intent-based slice orchestration.

- **Rejected.** Hype-driven, with no falsifiable evaluation possible without operator intents and data [67, 68].

5 Stage C — Weighted ranking

Scores run from 1 to 10 and use the brief's weights: novelty 25%, significance 20%, feasibility 20%, rigour 15%, validation 10%, fit 10%. They are computed reproducibly by `scripts/score.py` from `data/candidate_scores.csv`. **The weighted total summarises our judgement; it is not evidence of publishability.**

Weighted ranking. Hard-rejected candidates are placed below retained ones regardless of score.

Rank	ID	Nov.	Sig.	Feas.	Rig.	Val.	Fit	Weighted	Decision
1	C1	7	8	7	8	7	8	7.45	Shortlist (primary)
2	C3	6	7	7	8	6	7	6.80	

Rank	ID	Nov.	Sig.	Feas.	Rig.	Val.	Fit	Weighted	Decision
									Shortlist (backup)
3	C6	6	7	6	7	6	7	6.45	Shortlist
4	C10	4	7	6	6	6	6	5.70	Retained
5	C4	4	6	8	5	7	5	5.75	Rejected (overlap)
6	C2	3	6	8	6	7	5	5.65	Rejected (gap contradicted)
7	C8	5	8	2	6	3	7	5.15	Rejected (infrastructure)
8	C5	4	8	2	7	3	7	5.05	Rejected (prior work and infrastructure)
9	C9	3	5	7	5	5	4	4.80	Rejected (no contribution)
10	C7	4	7	2	6	3	6	4.60	Rejected (infrastructure and crowding)
11	C11	3	6	5	4	4	5	4.45	Rejected (unfalsifiable)

Score justifications for the shortlist.

- **C1, novelty 7 rather than 9.** The statistical mechanism (LTT) and the idea of risk-controlled action gating [35] both exist. Novelty must come from the network-specific structure: per-slice budgets, multi-action policies, cross-slice coupling, and counterfactual evaluation. That novelty is plausible but has not been confirmed against full texts of [1, 26, 36].
- **C1, feasibility 7.** An emulated core is standard practice, but the existing testbed has not been audited.
- **C3, rigour 8, novelty 6.** Strong theory is possible, but [70] may already subsume the core result.
- **C6, feasibility 6.** Public datasets exist, but the 5G link is indirect.

6 Stage D — Adversarial novelty audit

6.1 C1: risk-controlled slice-aware intrusion response

Five most threatening prior works. These were identified from abstracts; full-text comparison is pending.

1. CALIBURN [1]: regime-dependent CRC for alert thresholds under an FP budget.
2. FIRCE [2]: "intrusion response" plus conformal evaluation, but the response is retraining.
3. CORA and Conformal Policy Control [35, 36]: risk-controlled gating of actions.
4. Hammar & Stadler [37, 38]: optimal intrusion-response policies with threshold structure (IEEE TNSM).

5. SliceSecure and the sinkhole-slice idea [18]: per-slice attack impact.

A sixth, RBSID [26], is risk-calibrated Bayesian decisions and must be read.

Claim-by-claim novelty matrix for C1 (✓ covered, ~ partially, ✗ not covered, ? unknown without full text, — not applicable).

Claim	CALIBURN	FIRCE	CORA/ CPC	LTT	Ham-mar-Stadler	RRE	SliceSe-cure	Jaqen	Status of claim
K1 Finite-sample distribu-tion-free guarantee	✓ (alerts)	~	✓	✓	✗	✗	✗	✗	**Not nov-el alone**
K2 Guar-antee on service-level con-sequence of actions	✗?	✗	~ (gener-ic harm)	—	✗ (model-based cost)	✗	✗	✗	Plausibly novel [In-ference]
K3 Het-erogen-eous per-slice budgets	✗?	✗	✗	—	✗	✗	✗	✗	Plausibly novel; thin alone
K4 Multi-action, non-monotone policy cal-ibration	✗?	✗	~ (CPC)	✓ (tool)	✓ (not dis-tribution-free)	✓	✗	✗	Novel only in combina-tion with K2
K5 Cross-slice coupling via coun-terfactual replay or twin, with error bound	✗	✗	✗	✗	~ (emula-tion)	✗	✗	✗	**Core novelty candid-ate** [Hy-pothesis]
K6 Valida-tion on a sliced 5G core with measured SLA KPIs	✗	✗ (IoT)	✗	✗	✗ (IT)	✗	~ (detec-tion)	✗	Novel if the test-bed exists

Incremental-novelty risk. If K2–K5 reduce to "apply LTT to a new loss", reviewers will call the work an application paper. The contribution must therefore include:

- **(i)** a problem formulation in which collateral risk is defined against an oracle-mitigation counterfactual, so that it isolates harm due to detection and action error;
- **(ii)** a twin-in-the-loop calibration result that propagates surrogate error into the guarantee, with an empirical bound on that error;
- **(iii)** a quantified analysis of temporal dependence across episodes;
- **(iv)** evidence that alert-level control (K1 alone) fails to bound per-slice collateral risk, the key falsifiable claim.

Strongest likely reviewer objections.

1. "This is LTT applied to networking."

2. “The i.i.d.-episode assumption is unrealistic for traffic.”
3. “Emulated RAN, so no 5G claims.”
4. “The counterfactual replay is expensive; its fidelity is unproven.”
5. “The baselines are straw men.”

The plan in Section 8 answers each one.

6.2 C3: SLA-weighted multi-slice quickest detection

Threats. Multistream QCD with second-order optimality [70]; adaptive CUSUM for intrusions [69]; an unpublished multi-slice QCD simulation reporting cross-slice coherence effects [71]. Fewer than five threats were found because QCD-for-5G-slicing is sparsely published, but the mature general theory is itself the main threat.

Novelty matrix (summary). Heterogeneous false-alarm constraints per stream: unknown, possibly covered in the multichannel literature (not searched in depth). SLA-weighted delay: plausibly novel. Coupled streams due to shared resources: partially addressed in [71].

Objection. “It is a corollary of known multichannel CUSUM results.” **Verdict:** survives as the backup only.

6.3 C6: calibrated GNN digital twin

Threats. RouteNet-Fermi [62] and the NDT vision paper [63]. We have **not** searched specifically for conformal or uncertainty-aware GNN network models; this is a known gap, G-07. **Verdict:** survives provisionally, ranked third.

7 Stage E — Real-world feasibility audit

Feasibility audit.

Resource	Status	Evidence
Compute (this session)	4 vCPU, 15 GB RAM, no GPU, no LaTeX toolchain	**[Verified]** by inspection
Network egress (this session)	Search only; scholarly APIs and arXiv blocked	**[Verified]** by proxy logs
Prior manuscripts and code	Repository empty	**[Verified]**
5G-NIDD	Public (IEEE DataPort, DOI 10.21227/xtep-hv36)	**[Verified]**, record only
5GC PFCP dataset; SliceSecure dataset	Public on DataPort	**[Verified]**, record only; label-quality concern noted
5G-SliciNdd (team dataset)	Not provided	**[Unverified]**
Team 5G testbed (eMBB/URLLC/mMTC shaping)	Described in the brief, not inspected	**[Unverified]**: core, RAN, UPF, orchestration, isolation, and mitigation hooks all unknown
Open-source sliced-core scaffolding	Open5GS Docker with per-slice UPF [22]	**[Verified]** that the repository is referenced
RAN / SDR / RIC	None found	**[Unverified]**; treated as unavailable

Choice of validation method for C1. The claims concern core- and user-plane mitigation and the resulting per-slice KPIs. An **emulated sliced 5G core** (Open5GS or free5GC with UERANSIM, a per-slice UPF, and tc/nftables enforcement) can test them directly. Radio realism is not required, provided the claims are restricted to that scope, and the emulated RAN will be stated as a threat to validity. It is combined with **trace-driven replay** from 5G-NIDD and the PFCP dataset for scale and repeated splits. Pure simulation is insufficient, be-

cause the collateral loss depends on real queueing and enforcement behaviour. A physical RAN is not required to test the hypotheses. **No new equipment is essential.** Access to and an audit of the team testbed is the main dependency (G-01).

8 Stage F — Provisional recommendation

8.1 Primary (provisional): C1

1. **Title (working).** Certified Collateral-Risk Control for Slice-Aware Intrusion Response in 5G Networks.
2. **Target journal.** IEEE Transactions on Network and Service Management (TNSM). Its scope covers security management, closed-loop automation, and service assurance, and it has published the closest response work [37]. The alternative is IEEE TIFS if the theoretical component turns out to be stronger.
3. **Motivation and problem.** See C1 in Section 4.
4. **Closest work and remaining gap.** See Table 4.
5. **Formal hypothesis [Hypothesis].**
   - **H1 (validity):** under exchangeable calibration episodes, the calibrated policy keeps the realised per-slice collateral risk at or below α_s for every slice, with probability at least $1-\delta$.
   - **H2 (necessity):** a policy built on alert-level FPR control or pooled calibration, at the same nominal level, violates the URLLC collateral budget in a measurable fraction of splits.
   - **H3 (utility):** at equal certified collateral budgets, the method's residual attack impact is no worse than that of a model-based threshold response policy [37] tuned on the same data.
6. **Candidate contributions.**
   - **(a)** The formulation: oracle-counterfactual collateral risk per slice.
   - **(b)** Twin-in-the-loop LTT calibration with surrogate-error propagation.
   - **(c)** A dependence- and shift-aware extension with a monitored recalibration trigger.
   - **(d)** An open, reproducible sliced-core mitigation testbed and benchmark protocol.
7. **Methodology.** Detector-agnostic: supervised gradient boosting and unsupervised Kitsune [73]. A structured threshold-policy family over actions. LTT with Hoeffding–Bentkus p-values and fixed-sequence or graph-based FWER control [33].
8. **Formulation (preliminary).** Let $s \in \mathcal{S}$ index slices. Let e be an episode (a time window), $\mathcal{B}_s(e)$ the set of benign UEs of slice s, and π_λ a policy that maps detector scores and slice identity to actions in $\mathcal{A} = \{\varnothing, RL(\rho), Q, B\}$. With $V_u(e;\pi)$ the SLA-violation indicator of UE u under policy π, and $\pi^\star$ oracle mitigation, define

$$L_s^c(e;\lambda) = \frac{1}{|\mathcal{B}_s(e)|} \sum_{u \in \mathcal{B}_s(e)} \mathbb{1}\{V_u(e;\pi_\lambda) = 1,\ V_u(e;\pi^\star) = 0\}, \qquad R_s^c(\lambda) = \mathbb{E}_e[L_s^c(e;\lambda)].$$

   Let $R^a(\lambda)$ be the expected residual attack impact. The target is

$$\hat{\lambda} = \arg\min_{\lambda \in \hat{\Lambda}} \hat{R}^a(\lambda), \qquad \hat{\Lambda} = \{\lambda :\ H_{s,\lambda} : R_s^c(\lambda) > \alpha_s \text{ rejected } \forall s\},$$

   with FWER control at level δ over all pairs (s,λ), which yields $\Pr\{\max_s (R_s^c(\hat{\lambda}) - \alpha_s) \le 0\} \ge 1-\delta$. This follows from LTT [33] and is not claimed as novel. **Proposition to be proved [Hypothesis]:** if a twin $\tilde{L}$ satisfies $|\tilde{L}_s^c - L_s^c| \le \varepsilon_s$ with probability at least $1-\eta$, then calibrating at $\alpha_s - \varepsilon_s$ certifies the real system at α_s with probability at least $1-\delta-\eta$. **To be developed:** a degradation term for β-mixing episodes or for TV-bounded shift, following [28].

9. **Architecture.** Detector → score bus → policy engine (calibrated thresholds per slice and action) → enforcement at the UPF (N3/N6 nftables and tc per UE, keyed by TEID or IP; quarantine via re-association to a sinkhole S-NSSAI) → KPI telemetry → calibration store and drift monitor (for example OWAD-style [24]).
10. **Data and testbed.** 5G-NIDD and PFCP traces for the replay arm. Team testbed or Open5GS-based emulation for counterfactual episodes. Benign traffic profiles per slice (eMBB bulk, URLLC periodic small packets, mMTC sparse).
11. **Baselines.**
    - B0: no mitigation.
    - B1: fixed-threshold block.
    - B2: conformal FPR-controlled detector with block (CALIBURN or Bates-style).
    - B3: pooled, non-slice LTT.
    - B4: cost-sensitive Bayes rule.
    - B5: optimal-stopping threshold response (T-SPSA) [37].
    - B6: oracle (upper bound).

    Ablations: remove slice stratification; remove twin-error correction; single-action versus multi-action; vary calibration size n. **Metrics:** realised per-slice collateral risk; frequency with which the guarantee is violated across at least 200 random and temporal splits; residual attack bytes; URLLC p99 latency; eMBB goodput; decision latency.
12. **Limitations and threats to validity.**
    - Emulated RAN: no radio claims.
    - Exchangeability of episodes.
    - Limited attack diversity (volumetric floods and scans; PFCP).
    - Adaptive attackers who game the thresholds.
    - Twin fidelity.
13. **Feasibility.** Moderate. All software is open source; the main effort is the replay engine.
14. **Major risks and mitigations.**
    - The certified set may be empty at a realistic n → use fixed-sequence testing, coarser policy grids, or larger budgets for mMTC.
    - Replay may be too slow → use a surrogate twin with an ε bound.
    - A full read may show that a threat paper (CALIBURN, RBSID, CPC) already covers K2 → fall back to C3.
15. **Minimum evidence before submission.**
    - H1 holds empirically on both replay and emulation.
    - H2 is demonstrated, or the paper is re-scoped.
    - The proposition is proved.
    - Code and data are released.
    - A full-text novelty audit against all six threats is complete.

Why C1 over the alternatives. It is the only candidate that

- **(i)** survives the hard-rejection criteria with an identifiable contribution,
- **(ii)** reuses the team's assets (slice-stratified security, uncertainty, and the slice-differentiated testbed) without duplicating detection-level work, and
- **(iii)** can be tested with infrastructure that is plausibly available.

8.2 Backup (provisional): C3

SLA-weighted multi-slice quickest attack detection under heterogeneous false-alarm constraints. **Target:** IEEE TNSM, or IEEE Transactions on Signal and Information Processing

over Networks if the theory dominates. Activate it if the C1 full-text audit finds K2 or K5 already covered.

9 Stage G (outline) — Phased roadmap

The detailed roadmap is withheld until the direction is approved, as the brief requires. Gate criteria for the first two phases:

- **Phase 1 (novelty validation).** Obtain and read the full texts of the six C1 threats, plus targeted searches G-02 to G-07.
  - **Go** if K2 and K5 remain uncovered.
  - **Stop or switch to C3** otherwise.
- **Phase 2 (formulation).** Prove the twin-error proposition and specify the dependence model.
  - **Go** if the proof holds and the certified set is non-empty in a pilot replay of 5G-NIDD with n of 500 episodes or fewer.
- **Phases 3–8** follow the brief: baselines, implementation, experiments, ablation, writing, and audit.

Effort estimate (assumes one full-time researcher with access to the testbed): Phases 1–2 in 4–6 weeks; Phases 3–6 in 4–6 months. **[Hypothesis]**

10 Evidence and access gaps

Evidence and access gaps.

ID	Gap	Blocks	Action needed
G-01	Team testbed not inspected (core, RAN, UPF, isolation, mitigation hooks)	Stage E, C1 feasibility	Provide configs, topology, and logs, or grant access
G-02	Full text of CALIBURN (and its retitled version), RBSID, CPC, CORA	C1 novelty (K2, K4)	Allow arxiv.org in the network policy, or upload the PDFs
G-03	Full text of Hammar & Stadler TNSM 2022 / 2024 and TIFS 2025	C1 baselines and novelty	IEEE Xplore access
G-04	CALM-5G (arXiv 2610.17962) existence and content	C1 / C2 novelty	Resolve the identifier
G-05	DOIs for the 20 V-META and 11 PARTIAL entries	Bibliography integrity	Allow doi.org and Crossref
G-06	Team’s prior manuscripts (5G-SliciNdd, leakage audit, slice-stratified security)	Overlap rule	Upload them
G-07	Targeted search on conformal or uncertainty-aware GNN network twins; multichannel QCD with heterogeneous constraints	C6, C3 novelty	Phase 1
G-08	Whether C-PP-COAD used 5G-NIDD (sources conflict)	C2 rejection evidence (non-critical)	Full text

11 Conclusion

Within the stated verification limits, the evidence supports abandoning detection-level conformal false-alarm control as a primary topic. It supports pursuing **certified collateral-**

risk control for slice-aware intrusion response (C1) as a provisional primary direction, with **C3** as the backup. The checkpoint stops here, pending approval and the closure of gaps G-01 to G-03.

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